

CREDGUARD - MODEL RISK AUDIT REPORT

Automated Silent Failure Diagnostic & Governance Suite

SYSTEM SEVERITY: CRITICAL

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1. Compliance & Risk Diagnosis

Significant feature drift detected across multiple core underwriting variables. Model risk distribution has shifted materially, and specific borrower subgroups exhibit approval rate declines exceeding regulatory tolerance limits. Immediate remediation required under model governance guidelines.

2. Drift Impact Prioritization (DIS = PSI x SHAP)

Feature Name	PSI Score	Drift Impact Score	Priority
AMT_INCOME_TOTAL	0.4287	0.3104	CRITICAL
AMT_CREDIT	0.2321	0.1618	CRITICAL
REGION_RATING_CLIENT	0.4065	0.1516	CRITICAL
AMT_ANNUITY	0.1432	0.1418	HIGH
DAYS_EMPLOYED	0.0887	0.0887	HIGH

3. Demographic Subgroup Impact

Demographic Subgroup	Approval Shift	Fairness Flag
Short-term Employed	-23.9%	FAIRNESS RISK
Unstable Employment	-20.8%	FAIRNESS RISK
High Income	-18.7%	FAIRNESS RISK
Middle Income	-18.2%	FAIRNESS RISK
Long-term Employed	-12.5%	FAIRNESS RISK
Medium Risk Region	-12.2%	FAIRNESS RISK

4. Mandatory Action Items

- Retrain model on recent production data immediately
- Recalibrate credit decision threshold
- Review fairness metrics across affected demographic segments
- Escalate diagnostic report to Model Risk Management committee
- Log findings in PostgreSQL compliance audit trail for regulatory inspection